

Moksh Shah

(647) 913-6108 | moksh.be@gmail.com | Toronto, ON | linkedin.com/in/moksh-be | github.com/mokshshah18

EDUCATION

Toronto Metropolitan University

BSc (Hons), Computer Science

Sept 2021 – May 2026

Toronto, ON

EXPERIENCE

SEW-Eurodrive Canada

Software Engineer

Nov 2025 – Present

Vaughan, ON

- Served as **technical lead** for a **5-person** intern team: conducted **code reviews**, ran daily standups, assigned sprint tasks, and facilitated **agile** retrospectives and velocity meetings.
- Acted as primary **stakeholder liaison**, gathering requirements and translating business priorities into **technical roadmaps** and sprint priorities across concurrent workstreams.
- Implemented a **real-time** dynamic obstacle avoidance system for a **5,000-lb AGV** (10,000-lb payload), improving moving-object tracking accuracy over legacy static detection algorithms.
- Applied **linear algebra** for real-time **position estimation** and **trajectory prediction** of dynamic obstacles, enabling safe **path planning** in complex warehouse environments.
- Designed a specialized **LLM for code analysis** across **100,000+ lines of code**, delivering **sub-1-second** query responses via a hybrid **SQL/NoSQL semantic search** architecture.
- Architected a custom **Linux API** to capture, process, and persist **ROS2** log messages into a **SQL** database, enabling centralized logging across **distributed** robotics systems.

Toronto Metropolitan University – Autonomous Vehicles Lab

Research Assistant

Oct 2025 – Apr 2026

Toronto, ON

- Built orbital simulations in **Unreal Engine 5** using **C++**, applying **linear algebra** to model orbital trajectories; developed custom lighting and weather systems.
- Mentored** an undergraduate aerospace engineer through the completion of their **machine learning** thesis.

SOTI

Software Developer Intern

Sept 2023 – June 2024

Mississauga, ON

- Migrated product architecture from sequential **MDM** to parallel **DDM**, boosting response times by **70%** using **C#**, **TypeScript**, **AngularJS**, **HTML**, and **SCSS**.
- Optimized the **CI/CD pipeline** and resolved race conditions in legacy code, reducing build times from **8 hours to 30 minutes (93% reduction)**.
- Authored **unit tests** achieving **90%+ code coverage** using behavior-driven testing frameworks, significantly improving validation reliability.
- Took on junior PM duties: drafted **Jira** tickets, translated client requirements, and coordinated with the dev team.

Toronto Metropolitan University – Autonomous Vehicles Lab

Research Assistant

May 2023 – Sept 2023

Toronto, ON

- Led a team of **5** graduate aerospace engineers to build a **LiDAR-based SLAM** system for autonomous navigation on rovers and drones.
- Deployed **TensorFlow Lite** object detection models on **Raspberry Pi**, optimized via **semantic segmentation** for embedded performance constraints.
- Assembled a research rover with a custom **Linux** distribution using **C++** and **Python**; conducted workshops for **10+** graduate students on **Bash**, **Git**, **ROS2**, **CMake**, and **Gazebo** simulations.

Studio 98

Software Developer Intern

April 2022 – Aug 2022

Clearwater, FL

- Built the front-end framework for a business management app using **TypeScript** and **React Native**, delivering **cross-platform** iOS/Android compatibility and improving development efficiency by **50%**.
- Integrated **REST API** calls for backend data, built a login/form authentication layer, and engineered a modular file structure to enhance scalability and reduce concurrent rewrite risk.

AWARDS

University of Guelph CTF – 3rd Place (Nov 2024) | Splunk CTF – 3rd Place | Scotiabank Career Builder Program

TECHNICAL SKILLS

Languages: Python, C++, C, C#, Java, TypeScript, JavaScript, SQL, Kotlin, R, HTML/CSS, SCSS

Frameworks & Libraries: Node.js, Three.js, Flask, Django, REST APIs, AngularJS, Vue.js, Next.js, React Native, .NET, ASP.NET, JUnit, Selenium, NumPy, Pandas, OpenGL

Tools & Platforms: Git, Docker, Kubernetes, Jenkins, Linux/Unix, Ubuntu, Kali Linux, Jira, Firebase, Wireshark, Figma, Gazebo, ROS2

Concepts: Data Structures & Algorithms, OOP, Distributed Systems, Microservices, Concurrency, CI/CD, Agile/SCRUM, Unit Testing